

EPILEPSY

Key Family Physicians: Dr Tan Wei Beng / Dr Ang Lai Lai

Specialist Advisor: Dr Rahul Rathakrishnan (Senior Consultant, Division of Neurology, National University Hospital)

Acknowledgement for their inputs: NUHSP: Ms Lee Ting Yee

Updated in October 2024 by Dr Tan Wei Beng. Next review date: October 2027.

CONTENTS

Objectives	3
Background	3
Definition	3
Symptoms and Signs (Not Diagnostic Criteria)	4
Differential Diagnosis	4
Investigations (If Relevant)	4
Referral for Evaluation of First Seizure	5
Anti-Epileptic Drugs (AED)	5
Pharmacological Treatment	7
Advice to Patients and Care Givers	9
Women, Pregnancy and Lactation	10
Fitness Certification	10
Emergency Treatment of Seizures	11
Living with Epilepsy	12
Recommended Care Components for Epilepsy	12
References	12



OBJECTIVES

1. Appreciate the different presentation of epilepsy.
2. Understand how a diagnosis is established.
3. Be familiar with the common medications used in the treatment of epilepsy and their associated side effects.
4. Special consideration-the woman patient during pregnancy and lactation.
5. Be familiar with issues related to fitness certification for patients with epilepsy.
6. Be prepared for the emergency management of a patient during seizure.

BACKGROUND

1. Epilepsy is one of the more serious yet treatable neurological disorders, affecting over 50 million people worldwide.
2. An estimated 20 million new cases occur each year globally.
3. If properly treated, about 70 - 80% people with epilepsy could lead normal lives.

DEFINITION

1. Epilepsy is a chronic neurological disorder characterized by recurrent unprovoked seizures. The seizures can be partial or generalised.
2. They are divided into the following categories:
 - a) Partial (focal) seizures – simple or complex
 - b) Generalized seizure – absence (petit mal), tonic-clonic (grand mal), myoclonic, tonic, clonic, atonic

SYMPTOMS AND SIGNS (NOT DIAGNOSTIC CRITERIA)

1. Aura
2. Cyanosis
3. Loss of consciousness
4. Motor manifestations:
 - Generalised stiffness of body and limbs followed by jerking of limbs
 - Tongue biting
 - Urinary incontinence
5. Post-ictal:
 - Confusion
 - Muscle soreness
 - Headaches

DIFFERENTIAL DIAGNOSIS

With loss of consciousness:	With abnormal movement:
▪ Syncope ▪ Cardiac arrhythmia ▪ TIA (transient ischaemic attack) ▪ Hypoglycaemia ▪ Panic attacks	▪ Movement disorders during sleep and when awake ▪ paroxysmal choreoathetosis, dystonia or tremor ▪ Drop attacks and cataplexy

INVESTIGATIONS (IF RELEVANT)

- a) Full Blood Count
- b) Electrolytes, Urea, Creatinine and Glucose
- c) Serum calcium, magnesium
- d) Liver function tests
- e) Electrocardiogram
- f) Electroencephalogram (EEG)
- g) Imaging – CT / MRI head

Notes on EEG

1. Often useful in the diagnosis, classification and prognostication of epilepsy.
2. Performed to support a diagnosis of epilepsy in adults in whom the clinical history is suggestive.
3. Should be performed soon after the attack when a helpful result is more likely to emerge.

REFERRAL FOR EVALUATION OF FIRST SEIZURE

1. A neurologist should evaluate all individuals with a first-onset suspected seizure.
2. This is to ensure accurate and early diagnosis, and initiation of appropriate therapy.
3. The primary care doctor can follow-up thereafter.

ANTI-EPILEPTIC DRUGS (AED)

At NUP setting, any AED should not be initiated at NUP, such therapy is only stepped down from neurologists in the hospital, i.e., the patient's epilepsy status should be stabilised with the AED's. Should any patient who lost follow up and consider reinitiation of the medicine, such case should be considered carefully on case-to-case basis, referral back to neurologist for reinitiation is recommended as a rule.

Consideration for AED

- a) The risk of seizure recurrence
- b) The benefits of being on AED

Commonly Used AEDs

- a) Phenytoin, carbamazepine, sodium valproate, phenobarbitone.
- b) Newer AEDs such as gabapentin, lamotrigine, Topiramate and levetiracetam, could be added-on by the neurologist for the treatment of individuals who have suboptimal control.

Changing the Formulation or Brand of AED

- a) Is not recommended because different preparations may vary in bioavailability or have different pharmacokinetic profiles.
- b) Increased potential for reduced effect or excessive side effects.
- c) The patient should be followed up earlier initially.

Monitoring of AED Levels

- a) Routine checking of AED levels in primary care setting:
 - Is unnecessary
 - Not cost effective
- b) For information. Hospital specialists order AED levels is useful for:
 - Assessment of compliance to drug treatment and titration of drug dosages.
 - Assessment of symptoms due to possible AED drug toxicity
 - Titration of phenytoin dose (as small doses can lead to disproportionately large increase in serum concentration)

Breakthrough Seizures

- a) Increased risk of breakthrough seizures due to:
 - Non-compliance to AED, drug interactions with AED resulting in lower AED levels
 - Alcohol abuse, sleep deprivation, concurrent illness
- b) Patients with frequent breakthrough seizures should be referred back to a neurologist.

If the patient experiences frequent breakthrough seizures, further re-evaluation should be done at the tertiary hospital. AED levels may guide the careful titration of drug dose(s). It will be challenging for NUP colleagues to measure / interpret serum AED levels during our day-to-day practice. Our specialist colleagues can help to stabilise the patient before referring back to polyclinic for continuation of care.

Withdrawal of AED

Withdrawal of AEDs can be explored in some cases:

- a) At the end of at least a 2-year seizure-free period.
- b) After a discussion on the potential risks and benefits.
- c) Referral should be made to the specialist clinic as this is best done by a neurologist. The speed of AED withdrawal is variable, and it depends on type of AED used. It is best referred to the neurologist for suggesting methods of withdrawal of AED concerned.
- d) The risk of relapse after withdrawal of AED is approximately 25% at 1 year and 29% at 2 years.

PHARMACOLOGICAL TREATMENT

Anti-Epileptic Drugs	Dosages (for adults)	Max Dose	Adverse Drug Reactions	Contraindications / Precautions / Remarks
Phenytoin (S1) (Dilantin®) Strength per capsule: 30 mg, 100 mg \$^ (Contains Phenytoin Sodium/ 92% phenytoin) Syrup: 125mg/5ml \$^ (Contains Phenytoin Acid/ 100% phenytoin)	Maintenance: 200-400 mg daily, in 3-4 divided doses Titrate in 30mg steps every 7-10 days (titration of phenytoin should only be done at the SOC and not at NUP)	400 mg daily	CNS symptoms (concentration-dependent): Drowsiness, dizziness, lethargy, nystagmus, blurred vision, diplopia, ataxia, slurred speech, confusion, coma, mood changes Nausea, vomiting, hepatotoxicity Gingival hyperplasia, hypertrichosis, thickening of facial features Hyperglycaemia, folic acid deficiency, peripheral neuropathy, vitamin D deficiency, osteomalacia Systemic lupus erythematosus, fever, rash	Toxicity: CNS symptoms (dizziness, far lateral nystagmus, ataxia, drowsiness, confusion, diplopia, slurred speech); exacerbations of seizure frequency, hyperglycaemia Potentially fatal (rare): Severe Cutaneous Adverse Reactions (SJS/TEN) may occur 1-3 months after initiation), agranulocytosis, aplastic anemia (Dosage adjustments and closer monitoring may be necessary when switching between Dilantin® capsules and suspension) Drug interactions: Space at least 2 hours apart from enteral feeds
Carbamazepine (S1) (Tegretol®, Tegretol CR®) Strength per tablet: 200 mg \$^ 200 mg CR \$^	Maintenance: 800-1200 mg daily in divided doses	1600 mg daily, many get toxicity beyond this, some patients require 2400 mg daily	CNS symptoms: Drowsiness, dizziness, lethargy, ataxia, mood changes Nausea, vomiting, constipation, dry mouth, hepatotoxicity Hyponatremia/SIADH, heart failure, proteinuria, acute renal failure Hair loss, allergic skin reactions Blood disorders, leucopenia, thromboembolism, disseminated lupus erythematosus	Toxicity: CNS symptoms (dizziness, nystagmus, ataxia); AV blocks, arrhythmias Potentially Fatal (rare): Severe Cutaneous Adverse Reactions (SJS/TEN) may occur 1-3 months after initiation. Estimated incidence of 1 to 6 per 10,000 new users with mainly Caucasian populations, but the risk in some Asian countries is estimated to be about 10 times higher; agranulocytosis (in prolonged fever); aplastic anaemia, hepatic failure Obtain HLA-B*1502 genotyping test results prior to CBZ initiation.
Sodium Valproate (S1) (Epilim®, Epilim Chrono®) Syrup: 200 mg/5ml \$^ Strength per tablet: 200mg EC \$^ 200mg, 300mg, 500mg (Chrono) (CR) \$^	Maintenance: 1000-2000 mg daily in 1-2 divided doses	2500 mg daily	CNS symptoms: Drowsiness, dizziness, headache, lethargy, abnormal dreams, ataxia, confusion, amnesia, mood changes (anxiety, depression) GI disturbances (abdominal pain, nausea, anorexia, diarrhoea), hepatotoxicity, hepatobiliary disorders, hyperammonemia Transient hair loss, weight gain, amenorrhea, dysmenorrhea, gynaecomastia Vasculitis, thrombocytopenia, allergic skin reactions Fanconi-like syndrome (rare in children), peripheral oedema	Toxicity: CNS symptoms (dizziness, nystagmus, ataxia; drowsiness, confusion) Potentially Fatal: (case reports) Pancreatitis, severe hypersensitivity reactions with organ dysfunction

Anti-Epileptic Drugs	Dosages (for adults)	Max Dose	Adverse Drug Reactions	Contraindications / Precautions / Remarks
Phenobarbitone (S1) Strength per tablet: 10 mg, 60 mg \$^{\wedge}\$	60-250 mg daily at night		CNS symptoms: CNS depression or paradoxical excitation, drowsiness, insomnia, nightmares, impaired judgment, headache, hyperkinesia, ataxia, hallucinations, mood changes Nausea, vomiting, constipation Hypotension, bradycardia, syncope Allergic skin reactions Agranulocytosis, thrombocytopenia, megaloblastic anaemia	Toxicity: CNS symptoms (dizziness, nystagmus, ataxia, drowsiness, confusion), respiratory depression, decreased pupillary light reflex, tachycardia, bradycardia, hypotension Potentially Fatal (rare): Severe Cutaneous Adverse Reactions may occur 1-2 months after initiation
Gabapentin (S2) Strength per tablet: 100 mg, 300 mg \$-\$\$\$\$^{\wedge}\$	300-800mg tds		CNS symptoms: CNS depression, drowsiness, dizziness, ataxia, mood changes Peripheral oedema, weight gain	Exclusively (100%) cleared renally
Topiramate (S2) (Not available in NUP) Strength per tablet: 25mg, 50mg, 100mg	50-200 mg bd		CNS symptoms: Drowsiness, dizziness, cognitive dysfunction, word-finding difficulty, mood changes Metabolic acidosis, nephrolithiasis, angle-closure glaucoma, anorexia, weight loss, oligohidrosis	
Levetiracetam (S2) Strength per tablet: 500 mg \$-\$\$\$\$^{\wedge}\$	500-1500mg bd		CNS symptoms: Drowsiness, dizziness, fatigue, headache, mood changes (irritability, aggression) Increased BP	Potentially Fatal (rare): Severe Cutaneous Adverse Reactions may occur 1-2 months after initiation

[^] Cost \$ <\$10 per month \$\$ \$10 - <\$20 per month \$\$\$ \$20 - <\$30 per month
 \$\$\$\$ \$30 - <\$60 per month \$\$\$\$\$ \$60 - <\$90 per month \$\$\$\$\$\$ \$90 or more per month
 Amount payable depends on patient subsidy level and drug subsidy class (Standard Drug List S1 and S2, Non-Standard Drug NS)

ADVICE TO PATIENTS AND CARE GIVERS

Advice on Seizure Precautions

1. Situations associated with increased risk of breakthrough seizures
 - a) Non-compliance to antiepileptic medication
 - b) Drug interactions with antiepileptic medications resulting in lower efficacy of antiepileptic drugs
 - c) Alcohol misuse
 - d) Sleep deprivation
 - e) Concurrent illness

Advice on Seizure First-Aid

1. Place the patient in recovery position or on his / her side.
2. Remove surrounding objects that may harm the patient.
3. Do not place any object in the patient's mouth.
4. Call for an ambulance in the event of injury during seizure, prolonged seizure (> 5 minutes), or seizure clustering without return to patient's baseline state.

Home and Workplace Safety

1. Minimize exposure to open fires and sharp instruments. Consider use of microwave ovens and blenders.
2. Refrain from soaking in baths over extended periods of time or locking toilet doors. Showers should be preferred over baths.
3. Refrain from swimming alone.
4. Operation of heavy machinery is discouraged.

WOMEN, PREGNANCY AND LACTATION

1. Women with epilepsy should be referred to specialist care for preconception counselling and for follow-up if pregnant.
2. Monotherapy at the lowest AED dose to control seizures is recommended where possible.
3. Folic acid 5 mg per day should be given to women on AED from pre-conception till the first trimester of pregnancy to prevent neural tube defects.
4. AED is a not contraindication to breastfeeding. Women should be encouraged to breastfeed after discussion with their neurologist.

FITNESS CERTIFICATION

Patients with epilepsy may come for routine medical screening e.g. pre-employment, Outward Bound School (OBS).

Pre-Employment Medical Examination

1. Office-based and sedentary jobs should pose no increased risk to the patient.
2. Occupations that put the patient at risk should he / she have a seizure would not be suitable. These include operating heavy machinery or working at heights.
3. Refer the patient to a neurologist, designated factory doctor (DFD), or masters in occupational medicine [MMed(OM)] for pre-employment assessment in hazardous occupations.

Assessment for Fitness for Physical Activities

1. Physical activities for stable patients should not be restricted unless it posed a danger.
2. Activities and sports which involves heights, water, or aggressive physical contact should best be avoided.

Driving

1. The current Road Traffic Act may prohibit individuals with epilepsy from driving in Singapore.
2. Individuals must declare their epilepsy condition when applying for a driving license.
3. For individuals who have already obtained their driving licenses, it is their duty to inform the authorities should they develop epilepsy subsequently.

EMERGENCY TREATMENT OF SEIZURES

Protect the Patient

1. Remove hazards from the immediate surroundings.
2. Protect the patient from falling unsupported to the ground or striking objects.
3. Position the patient on their side, with the head supported in a neutral inline position.
4. Protect the head and other parts of the body from striking objects but do not restrain the patient.

Initial Assessment and Management

1. Establish Airway, Breathing and Circulation (ABC) and administer high concentration oxygen.
2. Check for hypoglycaemia.
3. Observe and record the pattern of the seizure(s).
4. Note and record the duration of the seizure(s).
5. Do not force anything, including your fingers, into the person's mouth.

Emergency Pharmacotherapy

1. This is required if seizures last 5 or more minutes or recur more than 3 times an hour.
2. Initial dose of 5-10 mg diazepam IV or rectally.
3. If no response, the same dose can be repeated after 10 minutes.
4. Monitor pulse rate, BP, respiratory rate, O₂ saturation closely for respiratory and circulatory effects.

Post-Treatment of Seizure

1. Patient should be sent to the Emergency Department for further treatment and evaluation.

LIVING WITH EPILEPSY

1. Many people diagnosed and treated for epilepsy are able to live full, active lives and many live seizure-free if they take medications on schedule.
2. Even individuals with uncontrolled seizures can make lifestyle adjustments to allow them to have a reasonable lifestyle.
3. There are many resources to help patients with epilepsy. A list of some sources are as follows:
 - a) Epilepsy Foundation, epilepsy.com
 - b) Epilepsy Institute, epilepsyinstitute.org
 - c) American Academy of Neurology, aan.com

RECOMMENDED CARE COMPONENTS FOR EPILEPSY

Recommended Care Components	Minimum Frequency*	Remarks
Seizure Frequency	Annually	
Seizure Type	Annually	
Seizure Free Duration	Annually	
Influenza Vaccination [#]	Annually or per season	As recommended under the National Adult Immunisation Schedule (NAIS) and National Childhood Immunisation Schedule (NCIS)

* *More frequently if clinically indicated*

[#] *This is a non-reportable clinical indicator*

REFERENCES

1. Chronic Disease Management Programme – Handbook for Healthcare Professionals. Ministry of Health Singapore 2022
2. Singapore Medical Association. Medical Guidelines on Fitness to Drive 2nd Edition, 2011
3. Philip N. Patsalos *et al.* Antiepileptic drugs - best practice guidelines for therapeutic drug monitoring: A position paper by the subcommission on therapeutic drug monitoring, ILAE Commission on Therapeutic Strategies. *Epilepsia*, 49(7):1239–1276, 2008
4. Drug References: Drugdex, Micromedex and Drug Info Handbook International from Lexi-Comp